PULSE-TWIN 2026 – System Architecture

An In-Silico Metabolic Digital Twin for Proactive Glycemic Forecasting & Clinical Intervention

Data Sources

Synthetic EHR (C-CDA)
- 108 patient records
- Diagnoses (SNOMED-CT)
- Medications (RxNorm)
- Lab results & vitals

Wearable Data (Empatica E4)
- Heart Rate
- Skin Temperature
- Electrodermal Activity (EDA)
- Activity & Sleep

(15,731 time-series readings)

CGM Data
- Glucose readings
- Trends & variability
- (Continuous stream)

Raw Data

Data Ingestion & Preprocessing

ETL Pipeline
- Parse C-CDA XML
- Load CSV (GlucoBench)
- Data validation

Data Cleaning
- Handle missing values
- Outlier detection
- Synchronization

Feature Engineering
- Time windows
- Lag features
- Physiological indicators

Unified Patient Dataset
- Structured + Time-series
- Patient-specific features

Clean & Structured Data

Patient Digital Twin

Personalized Patient Profile
- Demographics & Clinical History
- Insulin Sensitivity Factor (ISF)
- Carb Ratio (ICR)
- Baseline HbA1c, BMI, Comorbidities



- Metabolic Model
- Physiological State
- Behavior Patterns
- Risk Factors

Digital Twin Engine
- Patient-specific simulation
- Real-time state estimation
- Continuous model update

Twin State & Features

AI/ML Prediction & Simulation

XGBoost Regressor
- 30 min glucose forecast
- Feature importance
- R^2 = 0.987 (preliminary)
- RMSE = 1.43 mg/dL (preliminary)

Clinical Rules Engine
- ADA alert thresholds
- Hypo: <54 mg/dL
- Hyper: >180 mg/dL
- Rate-of-change detection

What-If Simulation
- Meal (carbs)
- Insulin (units)
- Exercise (duration/intensity)
- Stress (level)

Predictions, Alerts & Scenarios

Doctor Clinical Dashboard

- Patient Overview
- Glucose Forecast (30 min)
- Alerts & Notifications
- What-If Simulation
- Patient Twin Visualisation
- Clinical History & EHR



Infrastructure & Security

- Cloud / Local Server (Deployment)
- MySQL Database (Patient, EHR, Time-series)
- APIs (RESTful Endpoints)
- Authentication & Access Control (Role-based)
- Data Privacy & Encryption (HIPAA / GDPR compliant)

Technology Stack

- Python (3.10+)
- Flask (REST API)
- MySQL (8.0)
- XGBoost (2.0+)
- scikit-learn (1.4+)
- Pandas (1.5+)
- NumPy (1.26+)
- HTML5 / CSS3 / JS (Frontend)